

Scenario S3-O2-I1 — Our Level 2 (Ideal) / Investor Level 1 (Bootstrapped)

Scenario Summary

Compressed founder phase with constrained investor follow-on. The founders self-fund at ideal intensity (~100% of baseline spend) through Pre-Pilot and Pilot 1, building a credible prototype and live pilot with real patients over ~6 months. A bootstrapped-level investor arrives right after Pilot 1 (approximately M6-M7), providing minimum capital to bridge Pilot 2 and Production Readiness. The ideal-level self-funding produces a strong Pilot 1 result, but the bootstrapped investor capital creates resource tension during the scaling phases.

- **Economic cost to Production:** \$125,000-160,000 / INR 10,375,000-13,280,000
- **Target timeline to Production:** 24-30 months
- **Founder/company capital before investor:** \$53,000-65,000 / INR 4,399,000-5,395,000
- **Investor capital modeled:** \$72,000-95,000 / INR 5,976,000-7,885,000
- **Equity structure:** 60 / 20 / 20 (founder / founding team / investor pool)
- **Break-even month:** 30-36
- **Risk level:** High
- **Main risk to watch:** bootstrapped investor capital barely covers Pilot 2 + Production at constrained pace; strong Pilot 1 creates expectations that under-funded scaling cannot meet

Timeline

Phase	Timing	Duration	Minor Checkpoints	Gate to Advance
Pre-Pilot	M0-M4	4 months	Architecture freeze (M1), APK backlog lock (M2), doctor training draft (M3), admin callback SOP (M3), internal rehearsal cycles (M4).	Core web flows stable, internal rehearsals pass, doctor and admin runbooks versioned, APK build underway in parallel.
Pilot 1	M4-M6	2 months	Warm-network onboarding (M4), first paid consults (M5), callback calibration, doctor playbook v1 live, investor pitch deck finalized.	Ethiopia pilot runs live for 2 months with measured throughput; investor term sheet signed by M6.
Pilot 2	M6-M14	8 months	Marketing push (M8), partner reporting (M10), Kenya prep (M11), roster planning, partial automation, staffing calibration.	5k-10k user-capable operations with stronger APK, cleaner reporting, staffed callback coverage.
Production Readiness	M14-M24	10 months	Compliance closeout (M16), release freeze (M19), launch playbooks (M21), certification refreshers, production support schedule.	Current prototype scope is production-safe across web and patient mobile with documented fallback operations.

Feature and Output Checklist

Phase	Output Checklist
Pre-Pilot	Web core hardened at ideal pace with parallel Android build; doctor/admin training materials drafted; support desk scripts and callback trees defined; partner onboarding kits outlined. Sufficient concurrency for web and APK work to overlap.
Pilot 1	Patient app covers core journeys (registration, dashboard, consult); provider web is stable for GP and basic specialist flow; doctor roster, admin callbacks, and manual fulfilment visible in-system. Investor pitch deck with live data.
Pilot 2	Core patient features complete; partner dashboards functional at minimum viable level; staffing ladder active but lean; reporting and payment loops operational but manual-heavy. Kenya prep begins late in phase. Bootstrapped investor budget forces prioritization.
Production Readiness	Current prototype scope is production-grade across web and patient mobile. Trained ops staff with manual fallback available. Compliance documentation complete. Stretched timeline allows thorough but slow hardening.

Services and SLAs

Phase	Uptime Target	Response Commitment	Operating Model
Pre-Pilot	95.0%	Next business day	3 core team at ideal utilization + 7 part-time developers at ~480 hrs/mo + 2 advisors; no employed clinical coverage.
Pilot 1	97.0%	Same-day for critical issues	Minimum employed coverage: 2 doctors, 2 admin ops, 2 technical support. Partner clinics supplement.
Pilot 2	97.5%	4-hour P1 response	Minimum employed coverage: 3 doctors, 3 admin ops, 2 technical support. Constrained by bootstrapped investor capital.
Production	98.5%	2-hour P1 response	Minimum employed coverage: 5 doctors, 5 admin ops, 3 technical support. Below ideal ladder due to funding constraint.

Cost Breakdown

Summary

Category	Amount
Capital expenditure through Production	\$62,400 / INR 5,179,200
Operating expenditure before Production launch	\$85,100 / INR 7,063,300
Total economic spend to Production	\$147,500 / INR 12,242,500

Workstream Detail

Workstream	Economic Cost (USD)	Economic Cost (INR)	Notes
Product and founder office	\$37,400	INR 3,104,200	3 core team at \$20/hr x 88 hrs/mo x 6 months at ideal (M0-M6), then reduced utilization post-investor (M7-M24).
Web platform hardening	\$14,900	INR 1,236,700	745 hours at \$20/hr senior rate. Completed faster in Pre-Pilot/Pilot 1 at ideal pace.
Patient APK build	\$7,200	INR 597,600	Fixed Android quote: 480 hours at \$15/hr. Parallel with web in Pre-Pilot.
QA, DevOps, security	\$6,200	INR 514,600	~310 hours at \$20/hr. Reduced scope post-investor due to bootstrapped capital.
Admin and care operations	\$52,300	INR 4,340,900	Employed doctors, admin callback staff, and technical support across 20 months of live operations. Below-ideal ladder due to bootstrapped investor.
Partner onboarding and provider success	\$7,500	INR 622,500	Doctor/admin training protocols, SOP writing, partner reporting.
Legal, accounting, compliance	\$4,200	INR 348,600	Friendly-network pricing at economic value.
Infrastructure, phones, and tools	\$15,800	INR 1,311,400	Hosting, devices, telecom/call costs across 24 months. Starts at ideal, constrained post-investor.
Marketing, travel, rehearsals	\$6,000	INR 498,000	Moderate Pre-Pilot/Pilot 1, lean post-investor.
Contingency and buffer	\$20,000	INR 1,660,000	~14% buffer on total. Tight given bootstrapped investor capital.

Operational Staffing Cost by Phase

Phase	Duration	Monthly Ops Cost (USD)	Monthly Ops Cost (INR)	Phase Total (USD)
Pre-Pilot (M0-M4)	4 months	\$0	INR 0	\$0
Pilot 1 (M4-M6)	2 months	\$2,482	INR 206,006	\$4,964
Pilot 2 (M6-M14)	8 months	\$3,200	INR 265,600	\$25,600
Production Readiness (M14-M24)	10 months	\$5,100	INR 423,300	\$51,000
Total operational staffing				\$81,564

Infrastructure Cost by Phase

Phase	Duration	Monthly Infra (USD)	Monthly Infra (INR)	Phase Total (USD)
Pre-Pilot	4 months	\$120	INR 9,960	\$480
Pilot 1	2 months	\$395	INR 32,785	\$790
Pilot 2	8 months	\$450	INR 37,350	\$3,600
Production Readiness	10 months	\$750	INR 62,250	\$7,500
One-time equipment (phones, laptops)	—	—	—	\$1,000
Total infrastructure				\$13,370

Effort and Team

Function	Staffing Pattern	Notes
Core team	3 part-time members @ ~4 hrs/day each, ~100% effective utilization pre-investor	Single owner/founder + 2 founding members. Parallel task execution enabled.
Technical delivery (pre-investor, M0-M6)	7 part-time developers, modeled at ~480 productive hrs/mo	Ideal capacity. Web and APK run in parallel.
Technical delivery (post-investor, M7-M24)	7 part-time developers, modeled at ~200 productive hrs/mo	Bootstrapped investor capital forces ~40% utilization. Significant slowdown.
Advisory board	2 fractional advisors @ ~10-15 hrs/mo each	Finance, compliance, strategic review.
Employed care coverage	Pilot 1: 2 doctors / Pilot 2: 3 / Production: 5	Below ideal ladder. Partner clinics critical.
Employed admin ops	Pilot 1: 2 admins / Pilot 2: 3 / Production: 5	Callbacks, exception handling, manual fulfilment.
Technical support	Pilot 1: 2 / Pilot 2: 2 / Production: 3	L1/L2 issue triage and live-ops monitoring.

Monthly Team Economic Cost

Period	Dev Team (USD)	Dev Team (INR)	Notes
Pre-investor (M0-M6)	~\$14,830	~INR 1,230,890	100% of ideal rate card. Full parallel capacity.
Post-investor (M7-M24)	~\$5,930	~INR 492,190	~40% of full rate card. Bootstrapped investor constrains capacity.

Revenue and Pricing

Revenue Source	Pilot 1	Pilot 2	Production
GP consults	Subsidised, Ethiopia-first, ~\$2-3 per paid interaction	Full Ethiopia pricing + Kenya tests	Core revenue line
Specialist consults	Minimal	Grows slowly after referral flow stabilizes	Lower-volume margin contributor
Pharmacy commission	Small, admin-assisted	Repeatable, manual-heavy	Scales with partner conversion
Diagnostics commission	Small, admin-assisted	Improves with order flow consistency	Retention lever
Travel/care coordination	Rare, high-touch	Low-volume, higher-value	Strategic differentiator

Revenue Line	Ethiopia Benchmark	Kenya Benchmark	Notes
GP consult	\$3.5 / ETB 200	\$5 / KES 650	Starts subsidised in Pilot 1; reaches benchmark by late Production.
Specialist consult	\$9 / ETB 513	\$12 / KES 1,560	Pilot 2 onward.
Pharmacy take-rate	10% of order value	10% of order value	Pilot 1 onward with manual fulfilment.
Diagnostics take-rate	12% of order value	12% of order value	Pilot 1 onward with admin/lab coordination.
Travel/care coordination	\$10-15	\$10-15	Small volume, high-touch.
Subscription	Deferred	Deferred	Not modeled before meaningful scale.

CAC and Key Business Metrics

Metric	Pilot 1	Pilot 2	Production
CAC	\$8	\$14	\$20
Gross margin	38%	48%	58%
Monthly ARPU	\$1.8	\$2.8	\$4.5
Monthly churn	14%	11%	7%
LTV/CAC target	1.4x	2.2x	3.0x

Cash Flow and Runway

Phase Summary

Checkpoint	Ending Cash (USD)	Ending Cash (INR)	Approx. Runway Remaining	Trigger Threshold
Pre-Pilot (M4)	\$5,300	INR 439,900	1.5 months	Trigger contingency if cash drops below 2 months of current burn.
Pilot 1 (M6)	\$4,800	INR 398,400	0.9 months	Critical: investor close must happen by M6-M7.
Pilot 2 (M14)	\$38,200	INR 3,170,600	4.5 months	Trigger contingency if cash drops below 3 months of burn.
Production Readiness (M24)	\$26,800	INR 2,224,400	3.2 months	Trigger contingency if cash drops below 3 months of burn.

Monthly Cash Flow

Month	Phase	Founder Capital In	Investor In	Revenue In	Outflow	Net	Ending Cash
M1	Pre-Pilot	\$13,000	\$0	\$0	\$11,650	\$1,350	\$1,350
M2	Pre-Pilot	\$13,000	\$0	\$0	\$11,650	\$1,350	\$2,700
M3	Pre-Pilot	\$13,000	\$0	\$0	\$11,650	\$1,350	\$4,050
M4	Pre-Pilot	\$13,000	\$0	\$0	\$11,750	\$1,250	\$5,300
M5	Pilot 1	\$13,000	\$0	\$500	\$14,832	-\$1,332	\$3,968
M6	Pilot 1	\$13,000	\$0	\$900	\$14,832	-\$932	\$3,036
M7	Pilot 2	\$0	\$83,500	\$1,400	\$10,130	\$74,770	\$77,806
M8	Pilot 2	\$0	\$0	\$1,800	\$10,130	-\$8,330	\$69,476
M9	Pilot 2	\$0	\$0	\$2,200	\$10,130	-\$7,930	\$61,546
M10	Pilot 2	\$0	\$0	\$2,600	\$10,130	-\$7,530	\$54,016
M11	Pilot 2	\$0	\$0	\$3,000	\$10,130	-\$7,130	\$46,886
M12	Pilot 2	\$0	\$0	\$3,400	\$10,130	-\$6,730	\$40,156

M13	Pilot 2	\$0	\$0	\$3,800	\$10,130	-\$6,330	\$33,826
M14	Pilot 2	\$0	\$0	\$4,200	\$10,130	-\$5,930	\$27,896
M15	Prod. Readiness	\$0	\$0	\$4,700	\$12,580	-\$7,880	\$20,016
M16	Prod. Readiness	\$0	\$0	\$5,200	\$12,580	-\$7,380	\$12,636
M17	Prod. Readiness	\$0	\$0	\$5,800	\$12,580	-\$6,780	\$5,856
M18	Prod. Readiness	\$0	\$0	\$6,500	\$12,580	-\$6,080	-\$224
M19	Prod. Readiness	\$0	\$0	\$7,500	\$12,580	-\$5,080	-\$5,304
M20	Prod. Readiness	\$0	\$0	\$8,500	\$12,580	-\$4,080	-\$9,384
M21	Prod. Readiness	\$0	\$0	\$9,500	\$12,580	-\$3,080	-\$12,464
M22	Prod. Readiness	\$0	\$0	\$10,500	\$12,580	-\$2,080	-\$14,544
M23	Prod. Readiness	\$0	\$0	\$11,500	\$12,580	-\$1,080	-\$15,624
M24	Prod. Readiness	\$0	\$0	\$13,000	\$12,580	\$420	-\$15,204

Totals:

- Founder capital in: \$78,000 / INR 6,474,000 (M1-M6 at \$13,000/mo x 6 months)
- Investor capital in: \$83,500 / INR 6,930,500
- Cumulative revenue: ~\$106,500
- Cumulative outflow: ~\$263,200

WARNING: Cash goes negative at M18. This path requires either (a) a bridge funding injection, (b) faster revenue ramp than modeled, or (c) cutting ops staff. The bootstrapped investor capital is insufficient to reach Production Readiness without supplementary revenue hitting targets. This is the defining fragility of S3-O2-I1.

ROI and Break-Even

Metric	Value
Total capital invested (founder + investor)	\$161,500 / INR 13,404,500
Total economic cost	\$147,500 / INR 12,242,500
Sweat-equity gap (economic cost minus cash invested)	~\$0 (capital exceeds modeled economic cost; gap is absorbed by the negative cash position in later months)
Cumulative revenue at M24	~\$106,500 / INR 8,839,500
Monthly break-even (revenue >= outflow)	~M24 (month 24)
Cumulative break-even (total revenue >= total capital)	~M30-M36 (6-12 months post-Production)
Time to ROI (revenue exceeds all economic cost)	~M36-M42 (12-18 months post-Production)

Fundraising, Valuation, and Dilution

Item	Value
Founder/self-funded capital before investor	\$78,000 / INR 6,474,000
Investor round size	\$83,500 / INR 6,930,500
Recommended structure	SAFE or milestone-based bridge note
Modeled pre-money valuation	\$417,500
Modeled post-money valuation	\$501,000
Modeled investor dilution from the 20% pool	16.7%

Ownership Pool	Before Round	After Modeled Round	Notes
Owner / founder pool	60%	56.0%	Retained majority control.
Founding + strategic pool	20%	18.7%	Shared pool for founding members and strategic partners.
Investor pool	20% reserved	16.7% issued / 3.3% remaining	Drawn from the dedicated investor pool with minor proportional dilution.

Key Risks

Risk	Likelihood	Impact	Mitigation / Trigger
Cash exhaustion at M18	High	Critical	Revenue must ramp faster than modeled, or bridge capital must be secured by M15. Monitor monthly cash position against 3-month runway threshold.
Investor does not close by M7	Medium	Critical	Pilot 1 data must be compelling. Pre-negotiate term sheet during M5-M6. Fallback to emergency self-funding if needed.

Strong Pilot 1 creates scaling expectations bootstrapped budget cannot meet	High	High	Manage partner and user expectations explicitly. Communicate scaling constraints in investor conversations.
Team velocity cliff post-investor	High	High	Dev team drops from ~480 to ~200 hrs/mo when bootstrapped investor capital replaces ideal founder funding. Build maximum possible scope in M0-M6.
Manual ops overload with lean staff	Medium	High	Add admin capacity before feature expansion. Track callback backlog as release gate.
Regulatory delay (Ethiopia / Kenya)	Medium	High	Continue partner-license model. Count friendly counsel at economic value.
Partner attrition during extended Pilot 2	Medium	Medium	Keep engagement through regular check-ins and incremental value delivery during the 8-month Pilot 2.
Kenya market entry delayed	Medium	Medium	Focus on Ethiopia stability first. Kenya prep is a Pilot 2 activity, not a blocker.

Recommendation

S3-O2-I1 is a **high-risk combination** that creates an asymmetric trajectory: a strong, well-funded Pre-Pilot and Pilot 1 followed by a resource-constrained scaling phase. The ideal-level founder investment produces a credible Pilot 1, but the bootstrapped investor capital is insufficient to sustain the operational staffing ladder through Production Readiness. Cash goes negative by M18, creating a hard dependency on revenue ramp or bridge capital.

This path is viable but fragile. It should only be pursued if:

- 1. No higher-investment investor options are available after Pilot 1
- 2. The founding team is confident that revenue will ramp faster than conservative projections
- 3. A bridge funding mechanism (founder loan, revenue-based bridge, or second small round) is available as a safety net
- 4. The team accepts that the dev team will drop to ~40% utilization after the investor enters, significantly slowing feature development

If the revenue ramp is slower than modeled or the investor close slips past M7, this scenario fails. Consider S3-O2-I2 for meaningfully lower risk, or S3-O2-I3 for faster execution with proper resourcing.

Assumption Notes

- Android is modeled from the external quote: 480 total hours at \$15/hr average across the mobile team.
- Remaining web work is normalized to 150%-160% of Android effort, with an expected case of 745 hours.
- Salary benchmarks for employed coverage: INR 60,000 for doctors (\$720/mo), INR 18,000 for admin ops (\$217/mo), INR 25,000 for technical support (\$301/mo).
- Our L2 (ideal) spend is modeled at ~~400% of team utilization, yielding ~\$43,000/mo founder contribution pre-investor~~ (\$14,830 team cost minus some timing offsets).
- Investor L1 (bootstrapped) contribution sized at ~\$83,500, representing bare-minimum capital to attempt Pilot 2 + Production Readiness at constrained pace.
- All INR conversions at 1 USD = 83 INR per params.md.
- FX buffers of 10-15% applied to ETB and KES revenue projections.

End of S3-O2-I1 scenario. All assumptions trace to params.md v2.